AI-Powered Crop Disease Monitoring System

Progress Report

In fulfillment of the requirements for the

NU 302 R&D Project

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Submitted by

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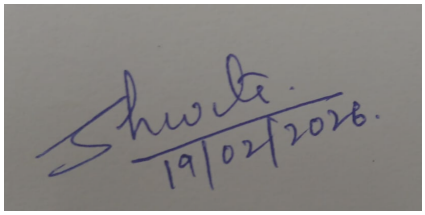
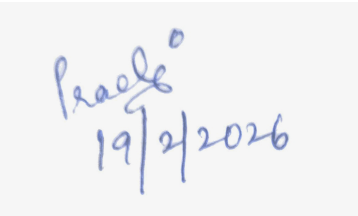
CERTIFICATE

This is to certify that the present research work entitled " AI-Powered Crop Disease Monitoring System" being submitted to NIIT University, Neemrana, Rajasthan, in the fulfillment of the requirements for the course at NIIT University, Neemrana, embodies authentic and faithful record of original research carried out by Adwik Pandey, Paarth Sehgal and Manav Kumar, student/s of B Tech (Computer Science and Engineering) at NIIT University, Neemrana,. She /He has worked under our supervision and that the matter embodied in this project work has not been submitted, in part or full, as a project report for any course of NIIT University, Neemrana or any other university.





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1. INTRODUCTION

Agricultural sector is a very crucial element of world food security and a major economic sector of the rural people in developing countries. However, in this sector, there is high productivity problem because the crops get affected by diseases that cause an annual loss in the yield that largely affects the stability of the economy. To reduce these losses, there is need to have an early and accurate identification of plant diseases in order to have sustainable agricultural development.

Historically, crop diseases have been detected through manual field inspection done by the farmers. Although such an approach is the norm, it is also becoming less reliable as it is highly reliant on human skills and subjective interpretation [4]. The timely or incorrect diagnosis of the disease can easily cause the further pathological development that cannot be stopped, and the yields are reduced, and the number of chemical interventions is also excessive. Such overuse of pesticides does not only escalate the costs of production but also expose environmental health and quality of crop to a lot of danger [8].

Automated crop disease detection has become a viable field of research that has improved greatly due to the implementation of artificial intelligence, especially through the deep learning technology. The Convolutional Neural Networks (CNNs) have proven to be very effective in the analysis of leaf images given their capability of isolating the complex visual patterns and diagnostic symptoms [6]. Recent research has followed different architectural optimizations, including the classical CNN comparisons with image segmentation [2] and advanced hybrid models (limiting local feature extraction of CNNs to the global structure of Transformers) [1, 7].

In spite of these developments, there is still a problem in detecting diseases just by using visual information. The environmental variables that are intrinsically related to pathogen development include the temperature, humidity, rainfall, and the type of soil [5]. The existing diagnostic systems usually encounter predictive constraints due to their inability to consider the individual climatic conditions that provide the best environments to various plant pathogens [3]. Moreover, most of the models are very accurate in laboratory conditions but they tend to perform poorly in the conditions of multi-crop or field environment with high vegetative noise [9].

There is thus an ardent requirement of smart systems that integrate visual data with environmental information to defeat the shortcoming of the existing diagnostic frameworks. This study is a proposed strong and dynamic method to overcome the difference between laboratory accuracy and level of reliability in the field.

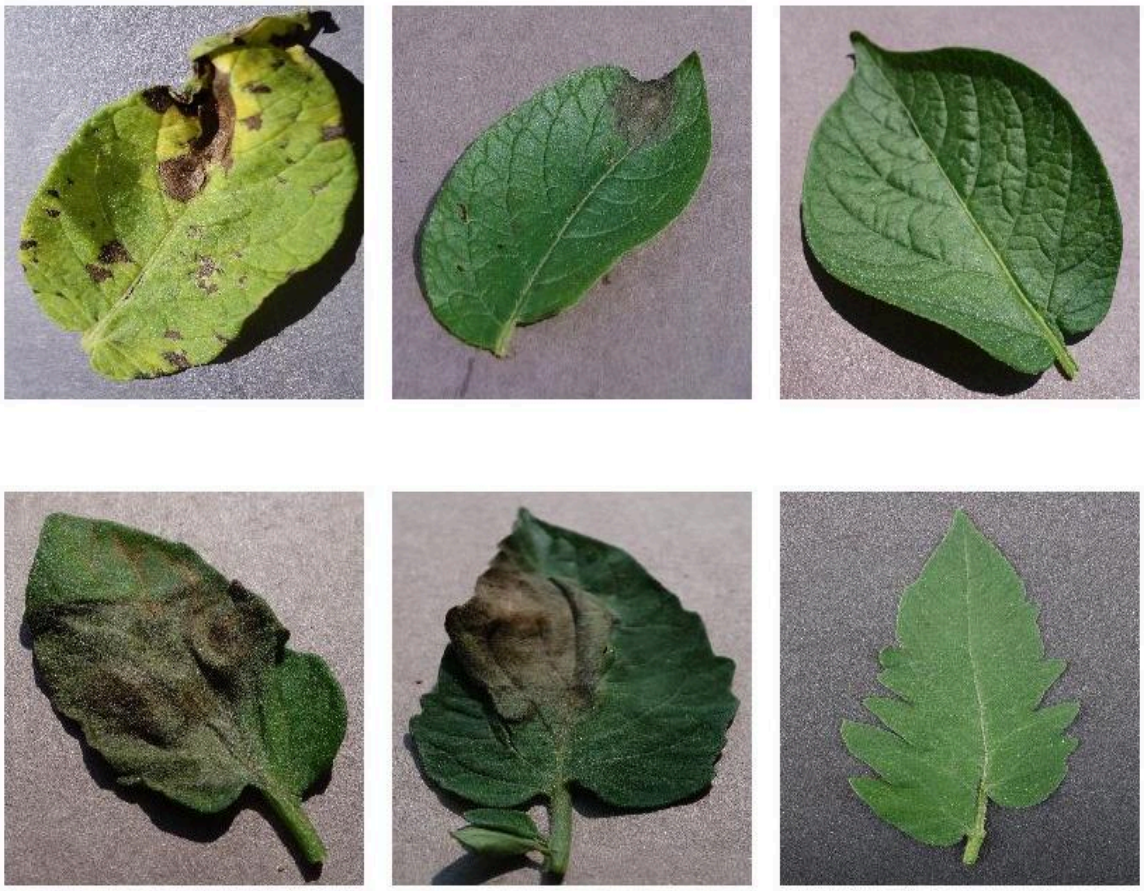

Figure 1 : The image shows potato (top row) and tomato (bottom row) leaves under three conditions: early blight, late blight, and healthy. Early blight leaves exhibit circular brown lesions with concentric rings, late blight leaves show irregular dark necrotic patches, and healthy leaves display uniform green coloration without visible symptoms.

2. LITERATURE REVIEW

S.No	Author(s)	Year	Title / Study Focus	Dataset Used	Method/Algorithm	Key Results
1.	Ayan Mondal, Ayan Chatterjee & Nurilla Avazov [1]	2026	A hybrid CNN-transformer model with adaptive activation function for potato leaf disease classification.	PlantVillage and Mendeley (Potato Leaf Disease)	PLDNet is a hybrid architecture using **DenseNet169** as a backbone combined with a **Transformer-based attention module** .	**99.54%** acc. on PlantVillage (Lab), **87.50%** acc. on Mendeley (Field); AFpM activation boosted acc. by **>3%**.
2.	Rekha R Nair, Ebin P. M., Tina Babu [2]	2024	CNN Architectures with Kmeans Segmentation and Data Augmentation	PlantVillage and Mendeley repositories	CNN Architecture including VGG16, VGG19 And ResNet50. For Preprocessing, K-Means clustering Segmentation [K=3,5,7]	VGG16 with K-means (K=3) and augmentation achieved **97.1% accuracy**
3.	Hyunseok Lee, Songho Yang , Tae-Jin Park[3]	2024	Multimodal crop disease diagnosis	Crop Disease Diagnosis dataset (4,675 samples)	ResNet50 CNN + LSTM with Multimodal Mixup Augmentation	**90.37% F1-score**; Multimodal data improved performance by 2.58% over images alone
4.	Md. Asif Iqbal, Kamrul Hasan Talukder [4]	2020	Detection of Potato Disease Using Image Segmentation and Machine Learning	Plant Village dataset	HSV color space masking/thresholding compared with SVM, KNN, Naive Bayes, etc..	Random Forest classifier achieved **97% accuracy**
5.	Avneet Kaur et al. [5]	2026	Crop disease surveillance through integration of machines and	Numerical dataset combining **Spore Trap Data and**	**ANN** Extracts synthetic features from complex weather data and **Random Forest**	**PEI:** 91%,**New Brunswick:** 86.5%, **Maine:** 84%, **Combined Dataset:** 87.5%

			deep learning in the face of climate change.	**Weather Station Data.**	Acting as the final binary classifier	
6.	Erlin, Indra Fuadi, Ramalia Noratama Putri, Dewi Nasien, Gusrianty, Dwi Oktarina20 [6]	2024	Evaluating 5 different CNN architectures to find the most effective model for classifying potato leaf diseases	PlantVillage dataset.	5 CNN architectures were pitted against each other: **VGG16**, **VGG19**, **MobileNetV2**, **ResNet50**, and **AlexNet**.	**ResNet50** achieved the highest performance rating with **97% testing**
7.	V. S. V. S. Murthy, G. N. Balaji & S. R. S. Reddy [7]	2026	*Plant Disease Detection using Hybrid Deep Learning with Architecture*	**PlantVillage :** A dataset containing **20,639** images of plant leaves. It covers **15 classes**,	**Hybrid VGG19-RF:** Uses VGG19 CNN for feature extraction and Random Forest (RF) for classification.	**95.6% Test Accuracy; 99.8% Train Accuracy. Precision, Recall, and F1-score $\approx$ 0.95.**
8.	Md. Samiul Islam,Saptak Roy Akash,Hayitov Abdulla Nurmatovich. [8]	2025	*PotatoCare: Automated detection and classification of potato diseases*	**10,117 images across 10 classes (tubers and leaves)**	**EfficientNetB3 with SMOTE for class balancing**	99.82% accuracy; 99.37% F1-score; high recall of 99.72%
9.	Sajjad Saleem, M. I. Sharif, M. I. Sharif, M. Z. Sajid, F. Marinello [10]	2024	*Comparison of DL models for Multi-crop leaf detection with vegetative feature isolation*	**Potato, Tomato, and Mango each having their healthy and disease counterpart**	**"AgirLeafNet" – A hybrid architecture combining NASNetMobile and Few-Shot Learning (FSL)**	**Potato:** 100% Accuracy.**Mango:** 99.8% Accuracy.**Tomato:** 92% Accuracy Fared better than traditional CNN.

Table 1 : Literature Review

3. KEY FINDINGS

When observed closely, the conclusion from all of these papers is that standard single models are no longer effective and sufficient. In order to get desirable results that actually work in real-life and not just laboratory testing, the shift is moving towards hybrid models that take two or models such as weather or simulated images. The 4 takeaway from the review is as follows;

1. *Hybrid is the new Standard;*

    - To address particular flaws, researchers are moving towards robust architecture [such as CNNs] and then further adhering to another algorithms [such as Transformers or RandomForest].
    - One of the Papers involves harmonizing CNN with Transformer and they have achieved **99.54% Accuracy.**

2. *Foundation still built on the ResNET and VGG;*

    - Researchers continue to rely on ResNET and VGG [both VGG16/19 as the 'dependable' basis for further usage of more intelligent tools and have been used for feature extraction.
    - One of the Papers achieved **97% Accuracy.**

3. *Data Augmentation is Necessary;*

    - Since real datasets are often too small or imbalance, researchers use Data Augmentation [may include scaling, shearing, translation etc] to increase the size of the dataset and enhance the training of the model to increase the accuracy.
    - One of the Papers quadrupled its dataset from 2,580 images to 10,320 images to reach the final accuracy of **97.1% Accuracy.**

The objective of the research work is to propose a dual-pathway diagnostic system for potato leaf diseases, offering farmers a choice between textual environmental data and image inputs.

4. GAPS AND CHALLENGES

The current state of AI-based plant pathology, especially in relation to potato plants, presents a number of critical impediments in progressing from laboratory prototypes to real-world deployable systems.

4.1. Dataset Limitations and Domain Shift

The most significant impediment in achieving real-time accuracy and reliability is the Lab-to-Field Domain Shift. Although PLDNet and EfficientNetV2B3-ViT perform near-flawlessly in laboratory conditions, their accuracy decreases by **12-15%** when applied in real-world conditions.

- **Environmental Bias:** Current CNN-based models are largely trained on "clean" data, i.e., centered leaves and uniform backgrounds. Another critical impediment is the lack of cross-validation, especially in relation to testing laboratory-based models against unstructured real-world data.
- **Long-Tailed Class Imbalance:** Current datasets, especially Mendeley, suffer from extreme class imbalance, e.g., 2% Nematode vs. 6% Healthy. Current techniques, especially SMOTE and augmentation, fail to deliver semantic complexity required by K-means segmentation in identifying rare classes.
- **Spectral and Temporal Narrowness:**Current research is limited to visible light spectra, i.e., RGB, and specific growth stages. Hyperspectral and multi-growth-stage analysis are critical in identifying asymptomatic infections during early growth stages.

4.2. Architectural Inefficiency and Edge Constraints

In order to deliver real-time diagnosis, the trade-off between network depth and speed remains an unsolved problem.

- **Computational Overload:** The models' high parameters (e.g., DenseNet-Transformer hybrids have around ~32M parameters) render them "unfit for the edge." The literature is flooded with accuracy results without any consideration of the models' FLOPs (Floating Point Operations) and the subsequent requirement for pruning.
- **Lack of Hardware-Aware Optimization:** The literature shows a systemic lack of employing any form of post-training optimization. INT8 quantization, TFLite conversion, and Knowledge Distillation are rarely practiced. The CNN-Transformer hybrids are computationally expensive without a corresponding increase in diagnostic accuracy.
- **Hybrid Volatility:**The CNN-Transformer hybrids show promise in incorporating contextual information but are also prone to instability. However, a standardized form of a lightweight hybrid for potato pathology detection has not been established.

4.3. Robustness and Statistical Rigor

The "accuracy saturation" on older datasets does not necessarily translate to robustness.

- **Morphological Mimicry & Occlusion:** The presence of complex factors such as leaf overcrowding, varying lighting conditions, and occlusions from other potato diseases results in a corresponding drop in the accuracy of the CNN-Transformer hybrids to around 85%.
- **Overfitting and Metric Inflation:** PlantVillage dataset accuracy rates exceeding 99% are likely a result of overfitting to artifacts. Moreover, there is a lack of statistical integrity in most literature. Standard Deviation (SD) and Confidence Intervals (CI) are rarely provided to evaluate the CNN comparison results.

5. PROPOSED METHODOLOGY

The proposed system architecture is divided into two specialized units, namely a Textual/Environmental Module with K-Nearest Neighbors (KNN) and a Visual/Imaging Module with the newly developed LitePLDNet. Such a "Choice-Based" technique can be diagnosed even in those cases when high-quality images are not available or when the environmental sensors form the main stream of information.

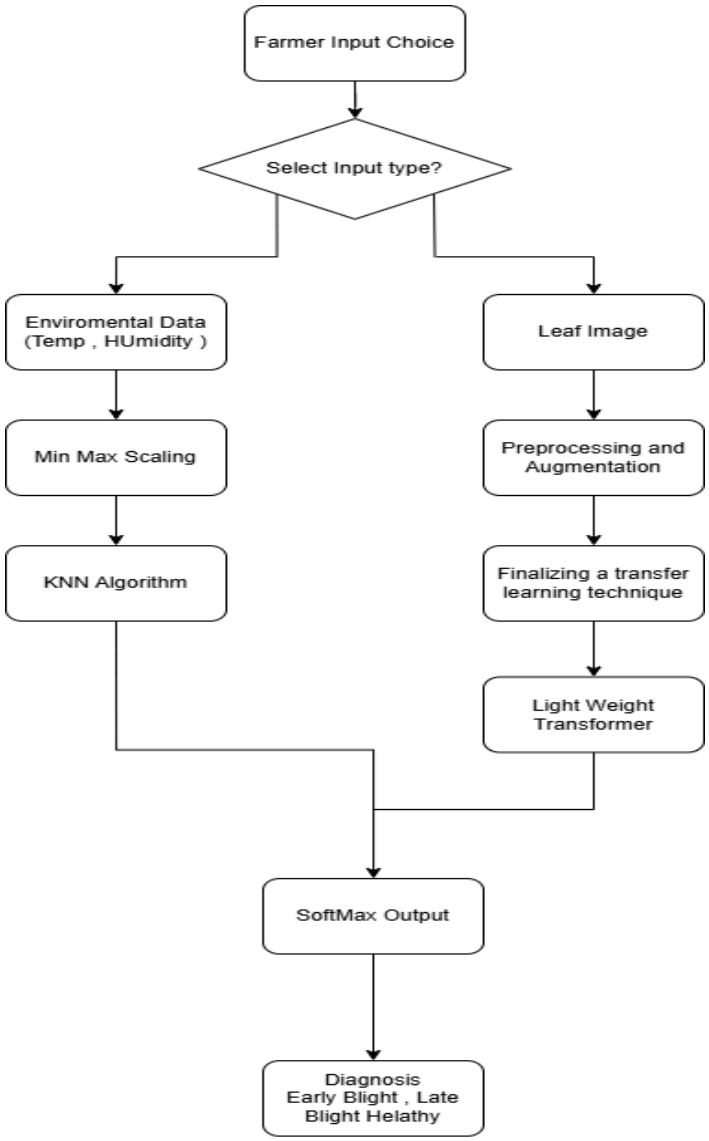

Figure 2 : Flowchart of the proposed methodology

5.1 Data Acquisition and Preprocessing

The framework utilizes a multi-modal data strategy to ensure high-fidelity inputs for both pathways.

- **Environmental Data [14]:** Measures of Temperature, Humidity, Wind Speed and Pressure are recorded. The preprocessing entails Min-Max scaling to normalize the features so that the distance based computation of the KNN algorithm is not skewed by the disparity of units of measurement.
- **Visual Data [12,13]:** A combination of the PlantVillage data[12] (controlled environment) and the Mendeley Potato data[13] (real field). Processing: The images are 224x224x3 and normalized. In order to fight field level noise and imbalance in classes, we use the MixUp (α=0.2) and CutMix (p=0.5) augmentations.

5.2 Pathway A: The Textual/Environmental Module (KNN)

A refined K-Nearest Neighbors (KNN) technique is proposed for farmers who provide textual or numeric sensor data.

- **Algorithm Rationale:** The reason behind the use of KNN is its high accuracy in tabular classification, as well as its lazy learning property, which enables it to learn the environment patterns without high retraining.
- **Operational Logic:** The input vector is mapped by the system into a multi dimensional feature space. The system classifies the disease (Early Blight, Late Blight, or Healthy) according to the majority of the *k* nearest cases in history (a simple majority vote is used).

Algorithm	Accuracy	Precision	Recall
KNN [K-Nearest Neighbour]	98.13%	0 [Late Blight] - 99% 1 [Early Blight] - 97%	0 [Late Blight] - 97% 1 [Early Blight] - 99%
SVM [Support Vector Machine]	97.88%	0 [Late Blight] - 99% 1 [Early Blight] - 97%	0 [Late Blight] - 99% 1 [Early Blight] - 97%
Decision Tree	95.14%	0 [Late Blight] - 96% 1 [Early Blight] - 95%	0 [Late Blight] - 94% 1 [Early Blight] - 96%
K-Means Clustering	87.66%	0 [Late Blight] - 77% 1 [Early Blight] - 98%	0 [Late Blight] - 77% 1 [Early Blight] - 98%

Table 2 : Comparison of Various Different ML Algorithms with their Result

5.3 Pathway B: The Visual Module (LitePLDNet Architecture):

The primary innovation under consideration for image-based diagnosis is **LitePLDNet**, a theoretical hybrid CNN-Transformer architecture. Based on our ongoing analysis of foundational literature, we propose a model optimized for edge-deployment on low-power devices like the Raspberry Pi 4.

6. TIMELINE

Week	***Date***	***Work Done***
Week-1	6th Jan 2026 to 12th Jan 2026	Base Paper + Basic Research
Week-2	13th Jan 2026 to 19th Jan 2026	Hybrid Model [Multi-crop Analysis]
Week-3	20th Jan 2026 to 26th Jan 2026	Collection of Datasets, Tabular [For ML Algorithms] Images [For Applying CNN]
Week-4	27th Jan 2026 to 2nd Feb 2026	Algorithms Tested, For ML [KNN, SVM, K-Means Clustering and Decision Trees] For DL [Convolutional Neural Network]
Week-5	3rd Feb 2026 to 9th Feb 2026	Testing Models + Approach Finalisation
Week-6	10th Feb 2026 to 16th Feb 2026	Proposing Methodology
Week-7	17th Feb to 19th Feb 2026	Documentation

Table 3 : Timeline

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